

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)

Assessment Tool Number	: CO3_AT2
Assessment Tool Weightage	: 35%
Assessment Tool	: Design Task
Duration	: 180 Minutes
Marks	: 100 Marks
Type of Assessment Conduction	: At Home
Evaluation Type	: GitHub
Result Generation	: Students design and justify appropriate wireless communication solutions by applying engineering principles to meet specified requirements.
Topic Covered	: Spectrum management, licensed/unlicensed spectrum, dynamic spectrum access, cognitive radio, OFDMA, resource block allocation, GFDMA, NOMA, power-domain multiplexing, SIC, and IoT applications. Advanced waveforms and wireless technologies including GFDM, FBMC, UFMC, mmWave propagation, path loss and penetration, millimeter-wave antenna systems, MIMO, and Massive MIMO fundamentals.

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills. Each student should answer (address) any 2 problem statements in Higher level as given below. Each question carries 50 marks

QUESTIONS: 2 (2 X 50 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)

2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)
19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)

39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

I S. Suresh (192512237) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

S. Suresh
Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

Introduction.

Remote and rural areas may not always have reliable terrestrial wireless Infrastructure because of geographical limitations, low population density, high deployment costs. A low Earth Orbit (LEO) satellite-assisted communication system can provide wireless connectivity to such locations.

LEO satellites operate much closer to the Earth than traditional geostationary satellites. Therefore they can provide relatively lower communication delay and support wide area coverage.

2. Objective.

The main objectives of the proposed LEO satellite assisted communication model.

- To provide wireless connectivity to remote regions.
- To extend communication coverage beyond terrestrial networks.

- To support voice, data IoT and emergency communication.

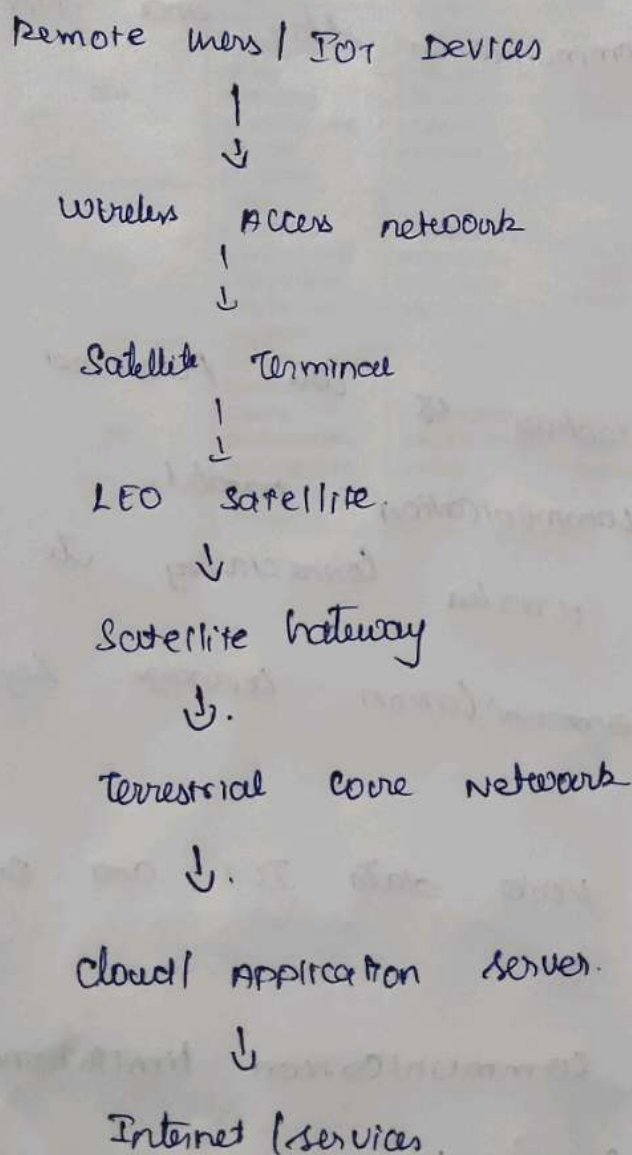
- To reduce communication limitations caused by geographical barriers.

Proposed System Architecture.

The proposed architecture consists of the following major components.

1. Remote user devices
2. wireless access network.
3. Satellite Terminal.
4. LEO satellite
5. Terrestrial / core network

Architecture Diagram.



working principal

The Communication process begins when a remote user generates data through a mobile device sensor or other wireless terminal.

The data is transmitted to a nearby wireless access point satellite terminal establishes communication with an overhead LEO satellite.

cloud/ Internet



core network



Satellite Gateway



LEO satellite



Remote Terminal



User Device.

5 Major Components:

a) Remote user Devices

These include smart phones, Computers, IoT devices, sensor, Emergency Communication equipment

And other wireless network

6 Communication flow.

User generates data



Wireless Communication



Satellite Terminal



LEO Satellite



Satellite Gateway



Terrestrial Core Network



Cloud / Internet



Destination Service

Conclusion.

The proposed LEO satellite-aided communication model provides an effective solution of extending wireless coverage to remote areas. The architecture connects remote users through satellite terminals and LEO satellites to terrestrial.

Introduction.

Modern Industrial environments contain machines that operate continuously and generate large amounts of operation data. Unconnected machine failures can cause production delays, increased maintenance costs and reduce productivity.

A smart factory Predictive maintenance system can use wireless sensors Industrial communication networks Edge computing and cloud based analysis to continuously monitor machine conditions.

Objective.

The objectives of the proposed Smart factory Information flow model are:

- To continuously monitor Industrial machines
- To collect machine operation parameters
- To detect abnormal operating conditions
- To analyze machine data using Edge / Cloud

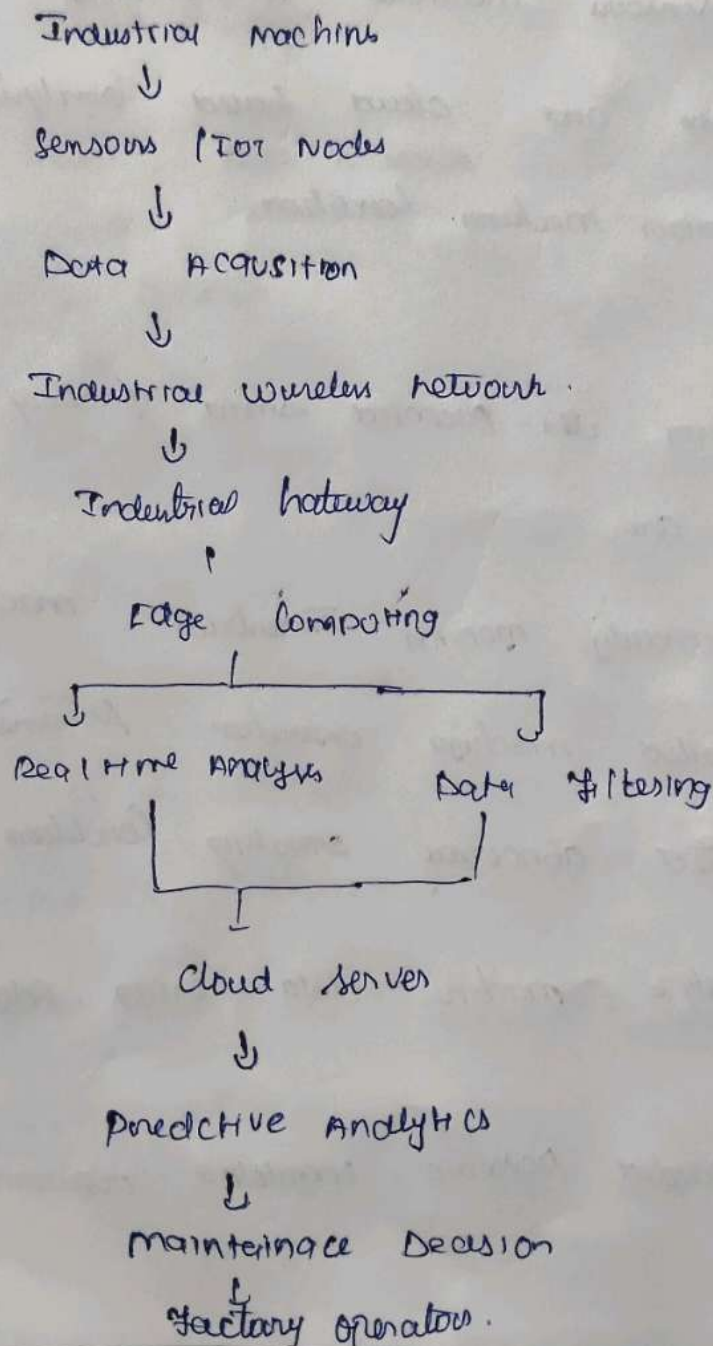
Processing

- To predict possible maintenance requirements

③ proposed Architecture.

1. Industrial machines
2. wireless / wired sensors
3. IOT sensor nodes
4. Industrial gateway
5. Edge Computing layer

Architecture Diagram.



Machines Data Collection

Sensors are Installed on Industrial equipment to monitor Important operating conditions

- Temperature
- vibration
- Pressure
- motor Condition
- machine speed
- current consumption
- operating time.

The sensors periodically continuously collect measurements and transmit them to Industrial communication network.

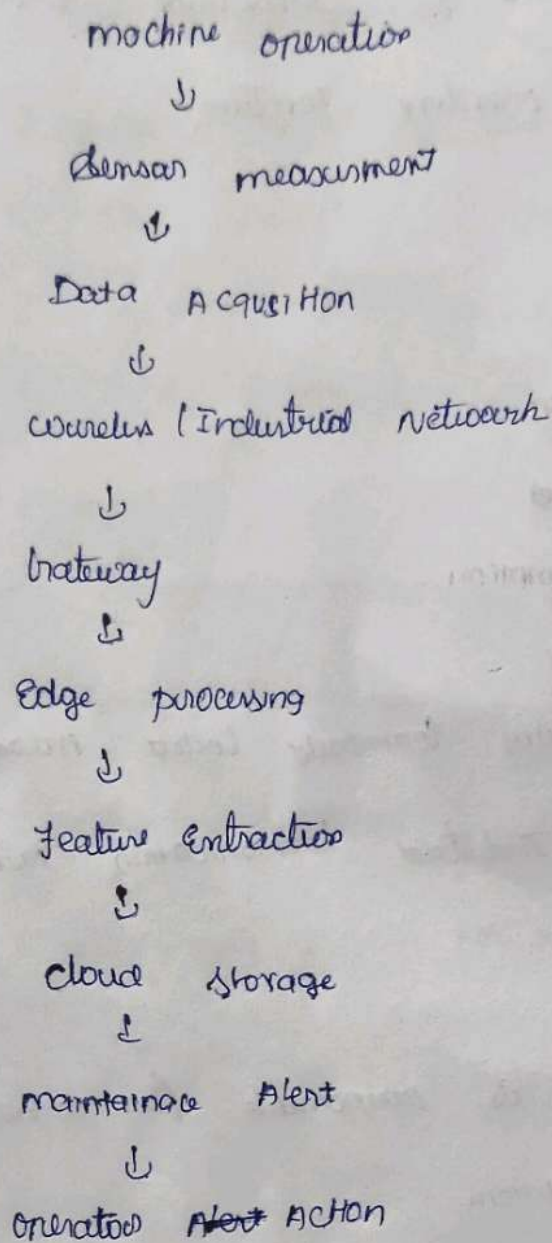
⑤ Sensor layer.

The sensor layer is responsible for observing machine operating conditions.

a vibration sensor can monitor abnormal vibration levels. A temperature sensor can detect excessive temperature.

The sensor nodes can perform basic preprocessing before transmitting the information which helps reduce unnecessary network traffic.

⑥ Information Flow



⑦ Conclusion

The proposed smart factory Information flow architecture provides a systematic approach for monitoring Industrial machines and supporting Predictive maintenance.